CIRP Journal of Manufacturing Science and Technology 2 (2010) 124–128

Contents lists available at ScienceDirect

CIRP Journal of Manufacturing Science and Technology

journal homepage: www.elsevier.com/locate/cirpj

Studies on hole-flanging process using multistage incremental forming

Z. Cui [a,b,*], L. Gao [a]

[a] College of Mechanical and Electrical Engineering, Nanjing University of Aeronautics and Astronautics, Nanjing 210016, PR China
[b] College of Engineering, University of Michigan, Ann Arbor 48109-2125, USA

ARTICLE INFO

Article history:
Available online 2 March 2010

Keywords:
Incremental forming
Hole-flanging process
Sheet metal forming
Thinning band

ABSTRACT

Hole-flanging is a manufacturing process used to produce flanges around holes in sheet metal parts. Traditional hole-flanging processes, which use dies and punches, are not cost competitive to make prototype parts or batches of small quantities. Since incremental forming does not require dedicated dies, it has shown promise to reduce cost and cycle time to manufacture such parts. In the present study, incremental forming with three forming strategies was investigated to produce prototype parts with hole-flanges. Results indicate that the forming strategy by increasing the part diameter in small steps during the forming process reaches the final optimum part geometry to improve the formability while it can produce a relatively higher neck height, maximum forming limit ratio (LFR_{max}), and uniform wall thickness. Also this strategy can be fine tuned to control the thinning band such that fracture can be avoided.

1. Introduction

High manufacturing costs and extended lead times to produce sheet metal prototype parts may pose financial and timing constraints on bringing new products having sheet metal components into the market. Also, cost and timing are critical issues when production of customized or small batches of sheet metal parts is required. Incremental forming, originally developed in Japan in the early 1990s [1,2], has shown promise to make such parts at a fraction of the cost and time compared to making the same parts using traditional forming technologies.

Incremental forming is a new type of a sheet metal forming technology which does not require dedicated sets of dies. On the contrary, a CNC controlled rigid tool with a smooth ball end tip is traversed or rolled on a sheet metal blank that is supported only at its edges to form the desired geometry. Similar to machining, the tool path determines the shape of formed product, since the tool is expected to cause only local deformation [3,4].

The benefits of incremental forming technologies are:

- Rapid and inexpensive manufacturing capability to produce prototypes and low volume components, due to the elimination of die construction and machining [5].
- The ability to form geometries more complex than that can be made using traditional stamping technologies.
- Shape and dimension accuracy can be controlled by changing the parameters such as pitch of tool path, tool diameter and forming speed.
- A higher forming limit can be achieved in incremental forming compared to traditional stamping processes [6,7].
- Incremental forming methods require only one clamping device and one forming tool, compared to the need to have multistage stamping dies and punches for different forming stages in traditional metal forming processes.

Hole-flanging process is one of the sheet metal forming processes that is widely used in automotive, aeronautics and astronautics industry. In conventional hole-flanging, a flat sheet with a pre-cut hole is formed as the punch moves down against a die in order to produce a smooth, round flanged lip with higher strength. The limiting forming ratio (LFR_{max}), which is defined as $d_p/d_{0\min}$, where d_p is the finished part diameter and $d_{0\min}$, denotes the minimum pre-cut hole diameter of blanks which can be successfully formed (see Fig. 1) [8–10]. Due to the need to make the punches and dies, this process is economical only for large volume production. Incremental forming of hole-flanging process does not need dedicated dies in order to produce parts with various geometries and, therefore, can produce parts at a lower cost than the conventional forming processes [5].

Previous work by Cui et al. [11] describes how incremental forming is used in hole-flanging process and the effect of various

* Corresponding author at: College of Engineering, University of Michigan, Ann Arbor 48109-2125, USA. Tel.: +1 313 522 6695.
E-mail addresses: cuizh2001@yahoo.cn (Z. Cui), meelgao@nuaa.edu.cn (L. Gao).

1755-5817/$ – see front matter © 2010 CIRP.
doi:10.1016/j.cirpj.2010.02.001

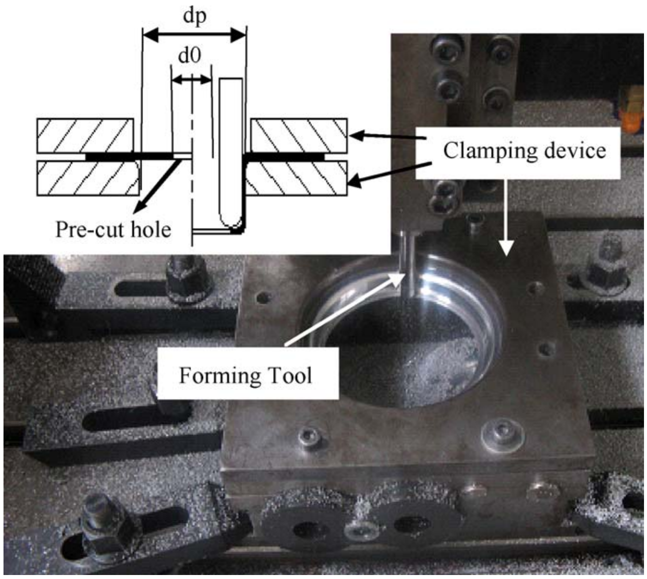

Fig. 1. Schematic of incremental hole-flanging.

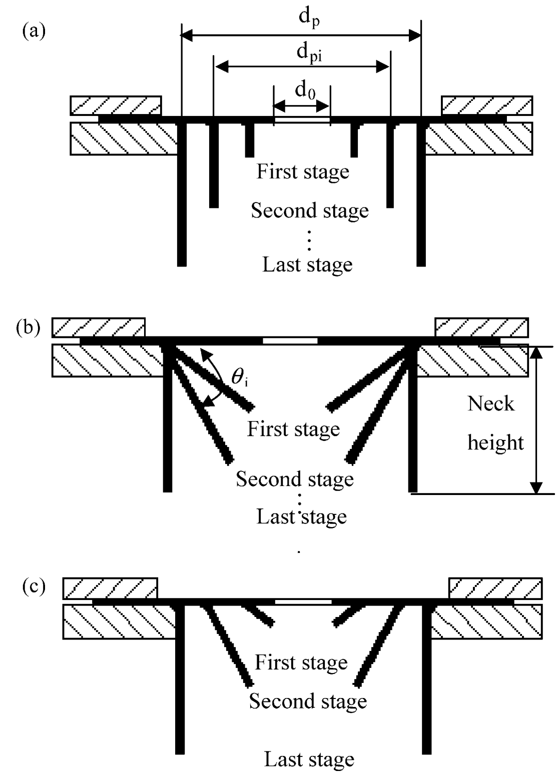

Fig. 2. Three tool paths are used in incremental hole-flanging process.

process variables in single stage incremental hole-flanging. As discussed in previous work, fracture will occur at the vertical lip wall because of the effects of thinning bands [12,13].

Multistage forming is considered by many researchers to improve the formability of incremental forming [14–17]. In the body of this work, three multistage tool path strategies are developed to optimize the formability of a hole flange. The thickness distribution, neck height and smallest pre-cut hole diameter of finished parts have been measured to determine which tool path strategy provides the optimum results.

2. Deformation schematic

Fig. 1 shows the schematic of the incremental hole-flanging process. A flat sheet with a pre-cut hole is clamped onto a forming fixture that is mounted onto the table of a three-axis CNC machine. The forming tool which is mounted to the spindle travels along a CNC tool path and incrementally forms the open area of the blank into a flange. 1060 aluminum sheet with 1 mm thickness is used as the experimental material with yield strength of 148 MPa, tensile strength of 151 MPa, and maximum elongation (engineering strain) of 13%.

3. Optimization of tool path in multistage incremental hole-flanging process

3.1. Tool path and experiment parameters

In order to develop and optimize a multistage incremental hole-flanging process, three forming strategies are attempted as shown in Fig. 2. With strategy (a), the pre-cut hole diameter d_0 and the blank size, which is selected on the basis of the final part diameter d_p, and wall angle (θ) are kept constant. The total amount of deformation is divided into a number of stages, as shown in Fig. 2 (a). From the initial stage to the final stage, d_{pi}, as defined in Fig. 2 (a), is increased to reach the desired diameter d_p. The reason to utilize multistage strategy (a) is that, material at the center is expected to generate more bending deformation instead of stretching deformation.

Table 1
Some experiment results for multistage incremental hole-flanging process.

Strategy	Stages	Smallest pre-cut hole diameter (mm)	Results	Neck height (mm)	LFR_{max}
Strategy (a)	2 (d_{pi} = 55, 65 mm)	28.5	Success	23.25	2.281
		26.5	Failure at wall	–	–
	3 (d_{pi} = 55, 60, 65 mm)	28.5	Success	19.51	2.281
		26.5	Failure at hole edge	–	–
	4 (d_{pi} = 50, 55, 60, 65 mm)	22.8	Success	18.64	2.851
		20.8	Failure at hole edge	–	–
	5 (d_{pi} = 45, 50, 55, 60, 65 mm)	22.8	Success	18.385	2.851
		20.8	Failure at hole edge	–	–
Strategy (b)	5 (θ_i = 30°, 45°, 60°, 75°, 90°)	30.5	Success	23.18	2.131
		28.5	Failure at wall	–	–
Strategy (c)	5 ($d_{pi} + \theta_i$ = 45 + 30°, 50 + 45°, 55 + 60°, 60 + 75°, 65 + 90°)	28.5	Success	21.225	2.281
		26.5	Failure at wall	–	–

With strategy (b) as shown in Fig. 2(b), the wall angle, is increased in steps to obtain the desired wall angle of 90° while pre-cut hole and all other geometrical parameters are kept constant.

Strategy (c) is the combination of the two previous strategies where both diameter d_{pi} and wall angle θ_i are increased in steps to obtain the desired wall angle 90° and desired part diameter d_p (see Fig. 2(c)). In all three strategies, no additional support is used underneath the work piece except for the forming fixture.

3.2. Experiments and results

A series of multistage experiments are carried out to find the smallest pre-cut hole diameter for each stage of strategy (a). A single stage can take up to 20 min to complete based on the part size. Other process parameters used during the experiments are: tool diameter (D) of 12 mm, vertical step size (Δ) of 0.5 mm, feed rate (f) of 2400 mm/min, and part diameter (d_p) of 65 mm.

As shown in Table 1, formability of a flat sheet with a smaller pre-cut hole can be improved by increasing the number of forming stages. The location where fracture occurs on the flange may change as a function of the number of stages. For example, fracture may occur near the forming fixture with two stages while it may occur near the pre-cut hole with three, four and five stages. Neck height of flanges without fractures with the smallest pre-cut hole diameter has been measured and the results show that neck height decreases as forming stage increases.

Only five stages of forming are needed with strategy (b) and (c) to achieve the smallest pre-cut hole. Part fractures may occur adjacent to the forming fixture in these two strategies and not at the pre-cut hole edge like in strategy (a). Strategy (b) has the highest neck height where as strategy (a) has the smallest neck height.

To get the optimum path strategy, sheet blanks with a pre-cut hole (d_0 = 35.5 mm) are formed using all three strategies with two, three, four and five stages. The vertical wall thickness is measured to determine how the thickness changed between stages and different strategies.

3.3. Discussion of three tool path strategies

3.3.1. Thickness of the flange formed by strategy (a)

Fig. 3 presents the wall thickness of the flanges formed using strategy (a). The sheet blank is formed in multi stages to obtain the

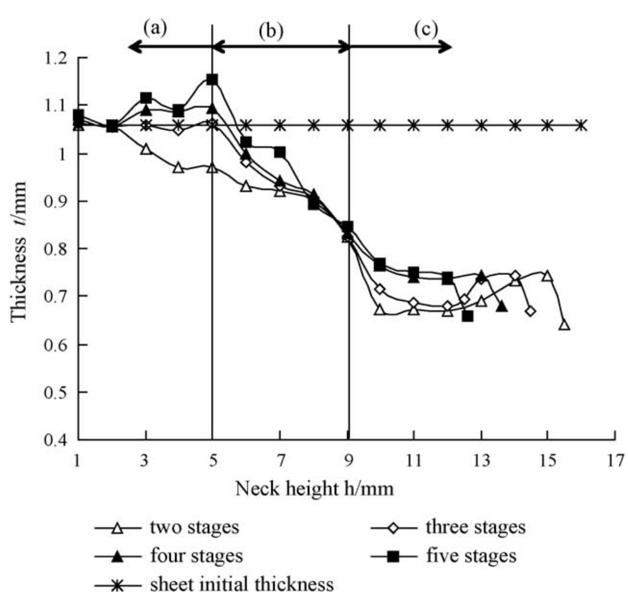

Fig. 3. The relationship of part thickness with neck height for the pre-cut hole diameter of 35.5 mm in strategy (a).

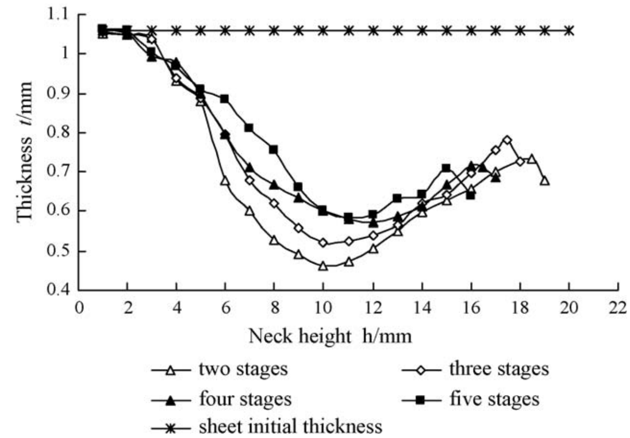

Fig. 4. The relationship of part thickness with neck height for the pre-cut hole diameter of 35.5 mm in strategy (b).

final part. As indicated in the figure, the final thickness of the finished part increases as the number of stages increases.

The thickness across the flange can be divided in to three regions. In region (a), the thickness is higher than the rest of the regions. This is due to bending near the clamped blank periphery. The thickness drops gradually after region (a), and again becomes stable in region (c). However, at the end of region (c), thickness again decreases. This indicates that the failure will occur at the pre-cut hole edge when the sheet thinning limit will surpass.

Fig. 3 also shows an increase in flange thickness with multistage forming compared to that of single stage forming. The reason is that, material at the center can be shifted to the outside [15]. A characteristic thinning band will occur on the formed wall near the forming fixture when the forming angle approaches to the maximum obtainable angle [13]. As shown in Fig. 3, there is no obvious thinning band with strategy (a), so it can be used as a means to avoid thinning band.

3.3.2. Thickness of the flanged parts formed by strategy (b)

Fig. 4 shows the wall thickness profiles of the same part formed with strategy (b). In comparison to Fig. 3, the characteristics of the thickness distribution of the parts are substantially different from those of the parts formed by strategy (a). The thickness drops to a minimum value and then start to increase again because of excessive sheet thinning, a phenomenon similar to necking, in the flange area. Part fracture

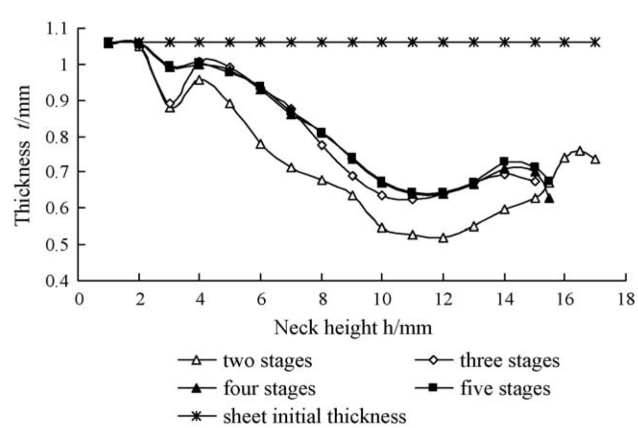

Fig. 5. The relationship of part thickness with neck height for the pre-cut hole diameter of 35.5 mm in strategy (c).

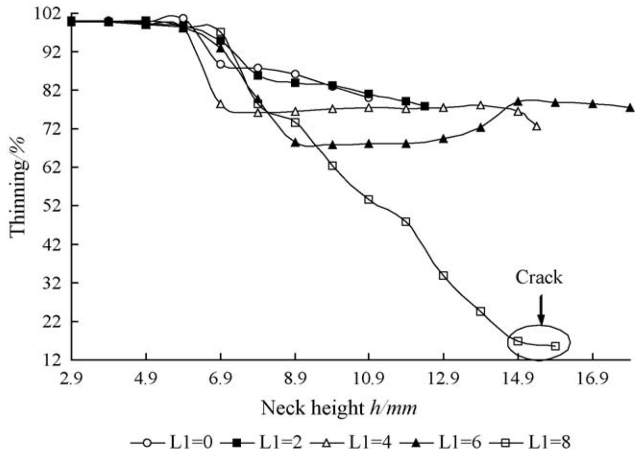

Fig. 6. The relationship of thinning and neck height with different $L1$.

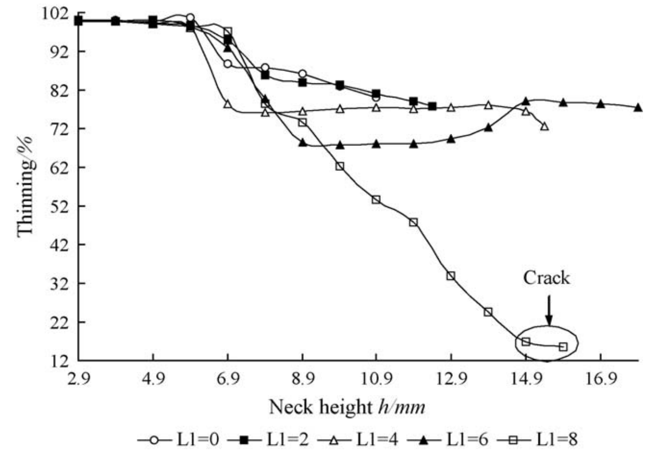

Fig. 7. The relationship of thinning and neck height with different combination of L while all of the finished parts diameters are 80 mm.

will occur at this location of minimum thickness after the thinning limit of sheet is surpassed.

3.3.3. Thickness of the flanged parts formed by strategy (c) and comparison with strategy (a) and (b)

As mentioned earlier, strategy (c) is a combination of the first two strategies. As can be seen from Fig. 5, wall thickness profiles of the parts formed with three, four and five stages are similar, while that the flange formed with two stages shows a considerable deviation from those mentioned above.

As is clear from above figures and Table 1, the relationship among the maximum forming limit ratios for the three strategies is

$$\mathrm{LFR}_{\max}^{\mathrm{strategy(a)}} > \mathrm{LFR}_{\max}^{\mathrm{strategy(c)}} > \mathrm{LFR}_{\max}^{\mathrm{strategy(b)}}$$

The thickness of the part formed by strategy (a) is the highest followed by that of the part formed by strategy (c). Also, flange thickness distribution is more uniform with strategy (a) than that with the other two strategies. Further, strategy (a) may avoid excessive thinning near the forming fixture. Therefore, it can be concluded that strategy (a) is the most versatile among the three forming strategies considered.

4. Optimization of forming stage using strategy (a)

To define the optimum number of forming stages with strategy (a), more experiments are conducted. Parts are formed with the following parameter values. D = 12 mm, Δ = 0.2 mm, f = 1800 mm/min, d_p = 80 mm, d_0 = 44 mm.

Li is defined as the horizontal distance between the center of the tool tip of the current stage and the hole edge, where i is the number of the stage.

The relationship of thinning and neck height with different $L1$ is shown in Fig. 6. It can be seen that, when $L1$ = 8 mm the part will fracture near the forming fixture. Thinning band will occur when $L1$ = 6 mm. There is no thinning band when $L1$ = 0, 2, 4 mm. This means that using $L1$ = 0, 2, 4 mm is a good choice in the first stage. Fig. 7 is the relationship of thinning and neck height with different combinations of L while all of the finished part diameter is 80 mm. It seems that $L1$ = 4 mm, $L2$ = 2 mm is the best combination. Because the part has a more uniform neck thickness and accepted neck height. Although $L1$ = 0 mm, $L2$ = 0 mm, $L3$ = 0 mm has the smallest thinning but the formed part also has the smallest neck height. So this combination is not an economical method.

5. Conclusions

In the body of work presented in this paper, incremental forming process is employed to produce flanged parts using three different forming strategies. Increasing the part diameter in small steps, named as strategy (a), to reach the final part geometry is proved to be an effective way to improve the formability of flanges. Strategy (a) can avoid thinning band phenomena effectively while strategy (b) and (c) have thinning band phenomena.

The best combination of parameters for strategy (a) seems to be $L1$ = 4 mm, $L2$ = 2 mm when used along with the current forming parameters. This means the distance *Li* is an important forming parameter in strategy (a). Based on the optimum value of *Li*, the formed part can be obtained with even neck thickness and relatively longer neck height.

References

[1] Matsubara, S., 1994, Incremental Backward Bulge Forming of a Sheet Metal With a Hemispherical-Head Tool—A Study of a Numerical Control Forming System, Journal of JSTP Plasticity and Processing, 35–406. 1311.
[2] Allwood, J.M., Utsunomiya, H., 2006, A Survey of Flexible Forming Processes in Japan, International Journal of Machine Tools & Manufacture, 46/15: 1939–1960.
[3] Iseki, H., 2001, An Approximate Deformation Analysis and FEM Analysis for the Incremental Bulging of Sheet Metal Using a Spherical Roller, Journal of Materials Processing Technology, 111/1–3: 150–154.
[4] Park, J.J., Kim, Y.H., 2003, Fundamental Studies on the Incremental Sheet Metal Forming Technique, Journal of Materials Processing Technology, 140/1–3: 447–453.
[5] Hirt, G., Junk, S., Bambach, M., Chouvalova, I., 2003, Process Limits and Material Behaviour in Incremental Sheet Forming with CNC-Tools, Materials Science Forum, 426–432:3825–3830.
[6] Allwood, J.M., Shouler, D.R., Tekkaya, A.E., 2007, The Increased Forming Limits of Incremental Sheet Forming Processes, Key Engineering Materials, 344:621–628.
[7] Jeswiet, J., Young, D., 2005, Forming Limit Diagrams for Single-Point Incremental Forming of Aluminium Sheet, Proceedings of the Institution of Mechanical Engineers Part B Journal of Engineering Manufacture, 219/4: 359–364.
[8] Leu, D.K., 1996, Finite-Element Simulation of Hole-Flanging Process of Circular Sheets of Anisotropic Materials, International Journal of Mechanical Sciences, 38/8–9: 917–933.
[9] Kumagai, T., Saiki, H., Meng, Y.G., 1999, Hole Flanging with Ironing of Two-Ply Thick Sheet Metals, Journal of Materials Processing Technology, 89-90:51–57.
[10] Huang, Y.M., Chien, K.H., 2001, Influence of the Punch Profile on the Limitation of Formability in the Hole-Flanging Process, Journal of Materials Processing Technology, 113/1–3: 720–724.
[11] Cui, Z., Gao, L., Lu, Q.J., 2008, Application of CNC Incremental Forming in Hole-Flanging Process, Chinese Journal of Materials for Mechanical Engineering, 32/7: 5–8.
[12] Hussain, G., Hayat, N., Gao, L., 2008, An Experimental Study on the Effect of Thinning Band on the Sheet Formability in Negative Incremental Forming, International Journal of Machine Tools & Manufacture, 48/10: 1170–1178.

[13] Young, D., Jeswiet, J., 2004, Wall Thickness Variations in Single-Point Incremental Forming, Proceedings of the Institution of Mechanical Engineers Part B Journal of Engineering Manufacture, 218/11: 1453–1459.
[14] Skjoedt, M., Bay, N., Endelt, B., Ingarao, G., 2008, Multi Stage Strategies for Single Point Incremental Forming of a Cup, International Journal of Material Forming, 1/Suppl. 1: 1199–1202.
[15] Verbert, J., Belkassem, B., Henrard, C., Habraken, A.M., Gu, J., Sol, H., Lauwers, B., Duflou, J.R., 2008, Multi-Step Toolpath Approach to Overcome Forming Limitations in Single Point Incremental Forming, International Journal of Material Forming, 1/Suppl. 1: 1203–1206.
[16] Hirt, G., Ames, J., Bambach, M., Kopp, R., 2004, Forming Strategies and Process Modeling for CNC Incremental Sheet Forming, CIRP Annals—Manufacturing Technology, 53/1: 203–206.
[17] Duflou, J.R., Verbert, J., Belkassem, B., Gu, J., Sol, H., Henrard, C., Habraken, A.M., 2008, Process Window Enhancement for Single Point Incremental Forming Through Multi-Step Toolpaths, CIRP Annals—Manufacturing Technology, 57/1: 253–256.